Md Mehedi Hasan Naeem

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GitHub | **LinkedIn** | **Google Scholar** | **ORCID** | **Portfolio**

RESEARCH PROFILE

Computer Science and Engineering undergraduate and Research Assistant focused on **Computer Vision, Pattern Recognition, Machine Learning, Visual Tracking, and Real-Time/Privacy-Preserving Intelligent Systems**. Contributed to **seven IEEE conference papers, six as first author**: one is published in IEEE Xplore, two have been presented and are in the proceedings publication stage, and four are accepted for upcoming IEEE conferences. A journal manuscript is currently under review at *Machine Vision and Applications* (Springer Nature).

EDUCATION

Jatiya Kabi Kazi Nazrul Islam University — Mymensingh, Bangladesh
Bachelor of Science in Computer Science and Engineering — Feb. 2023 – Present
CGPA: 3.44/4.00

Relevant Coursework: Machine Learning, Artificial Intelligence, Natural Language Processing, Data Structures and Algorithms, Computer Networks, Computer Graphics, Digital Signal Processing, and Computer Architecture.

RESEARCH EXPERIENCE

Research Assistant, Department of Computer Science and Engineering, JKKNIU Dec. 2025 – Present
Conduct research under Prof. Dr. Md. Sujan Ali in machine learning and intelligent systems. Work includes literature review, dataset preparation, experiment design, model evaluation, result analysis, and academic writing. Published three conference papers under his supervision; a journal manuscript is currently under review.

PEER-REVIEWED CONFERENCE PAPERS

Publications: all publications can be found here

Published in IEEE Xplore

- **MMH Naeem**, K Ruma, S Sultana, N Anjum, M Barua, and MA Islam, **"Solar-Powered IoT-Based Smart Farming Model (SPISFM),"** IEEE International Conference on Sustainable Technology and Engineering (i-COSTE), 2025. **Published in IEEE Xplore.** DOI: 10.1109/i-COSTE68047.2025.11467551.

Presented; Proceedings Publication in Progress

- **MMH Naeem**, K Ruma, N Anjum, S Sultana, and MS Ali, **"LUMO (Lightweight Unified Multilingual Orchestrator): A Privacy-Preserving Offline Voice Assistant,"** 3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026). **Presented; proceedings publication in progress.**
- **MMH Naeem**, MA Islam, M Barua, IA Araf, and MAH Mahir, **"A Machine Learning Framework for Predicting Restaurant Food Waste to Support Sustainable Food Management,"** 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026). **Presented; proceedings publication in progress.**

Accepted for Upcoming IEEE Conference Presentation

- **MMH Naeem**, K Ruma, N Anjum, S Sultana, MS Ali, and NJ Nemu, **"TheatreStageAI: Multi-View Retrieval-Augmented Generation for Structured Bengali Theatre Production Planning,"** IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2026). **Accepted; presentation scheduled.**
- **MMH Naeem**, TK Saha, IA Araf, and MAH Mahir, **"An Efficient Approach to Beetle Damage Detection of Banana Using Deep Learning,"** IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2026). **Accepted; presentation scheduled.**
- **MMH Naeem**, K Ruma, N Anjum, S Sultana, and MS Ali, **"A Hybrid Attendance Monitoring Framework Using Fingerprint Authentication and FaceNet-SVM Recognition,"** IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2026). **Accepted; presentation scheduled.**
- M Barua, A Islam, **MMH Naeem**, S Sultana, MAO Babu, and R Shaharin, **"TCN+LightGBM: A Hybrid Deep Learning–Gradient Boosting Architecture for Interpretable Short-Term Electricity Demand Forecasting in Bangladesh,"** IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2026). **Accepted; presentation scheduled.**

SELECTED RESEARCH & DEVELOPMENT PROJECTS

SmartPresence – Vision-Based Identity and Attendance Monitoring: Developed a computer-vision attendance framework using **FaceNet embeddings and SVM** for identity recognition, with **YOLO11n and ByteTrack** for person detection and tracking. The experimental setup includes 40 participants and focuses on reliable identity monitoring under real classroom conditions. GitHub

LUMO – Privacy-Preserving Offline Multilingual AI Assistant: Developed a lightweight multilingual voice-assistant framework designed to reduce dependence on cloud services through offline speech processing, local AI inference, and privacy-aware execution on resource-constrained devices. GitHub / Dataset

Deep Learning for Banana Beetle-Damage Detection: Worked on image-dataset preparation and deep-learning-based visual detection for agricultural damage assessment, with emphasis on practical computer-vision deployment and image-quality variation.

Smart Campus, Smart Transport: Developed a **Django-based transport platform** integrating **ESP32, GSM, and GPS** for real-time vehicle location tracking, routes, schedules, seat booking, and role-based services. GitHub | Live Site

JKKNIU Research Society Web Platform: Developed and currently maintain the society's **Django + React** platform, including event and executive management, deployment, custom-domain configuration, content updates, and ongoing technical maintenance. GitHub | Live Site

JKKNIU Model United Nations Club Web Platform: Led the development and continue to maintain the club's **Django + React** platform, including deployment, website updates, event and executive information management, and ongoing technical support. GitHub | Live Site

LEADERSHIP, TEACHING & SERVICE

President, JKKNIU Research Society (2026–Present): Lead an 18-member executive team and coordinate research programs, academic events, workshops, collaborations, and institutional communication. Currently overseeing an **ongoing two-month research training and workshop series** engaging approximately **200 students** with support from around **10 mentors**, covering research methodology, academic writing, publication preparation, higher study, and related research skills. In parallel, I am leading the initial planning and development of an **international research conference**, including sponsor outreach, guest and speaker coordination, partnership discussions, program planning, logistics, and other organizational requirements. Previously served as Vice President.

Founder, Amader Online School (2021–Present): Founded a free educational initiative and created 150+ learning resources; also teach and mentor junior learners in programming and problem solving.

Domain Leader, Aspire: Support AI and technology-focused member engagement and mentor-session coordination within the Aspire community.

Class Representative (2.5 Years): Coordinated communication between students and faculty, including academic updates, schedules, student concerns, and departmental matters.

Programming Trainer, Mid-Day Programming: Teach programming fundamentals and problem-solving skills to junior students through practical coding sessions.

SELECTED ACHIEVEMENTS & AWARDS

- ▷ **2nd Runner-Up, Idea Competition – Research Week 2026**, Research and Extension Center, JKKNIU.
- ▷ **1st Runner-Up, TechMind Hackathon 2024**, Department of CSE, JKKNIU.
- ▷ **Top 10 among 200+ Teams, Innovatex Hackathon**, BUBT IT Club.
- ▷ **Aspire Leaders Program 2024**, completed 30 hours of leadership and professional-development coursework.
- ▷ **Intel Student Ambassador – Level 1, 2025**.

CONFERENCE PRESENTATIONS & PARTICIPATION

- ▷ **11th IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2025)** – Research Presenter; Alexandria, Virginia, USA.
- ▷ **13th International BILTEK Congress** – Research Presenter; Paris, France.
- ▷ **MEETCON-WORK International Food, Agriculture and Veterinary Sciences Congress** – Research Presenter; Şanlıurfa, Türkiye.
- ▷ **12th International Cukurova Agriculture and Veterinary Congress 2026** – Research Presenter; Adana, Türkiye.
- ▷ **5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)** – Research Presenter; Bangladesh Army University of Engineering & Technology (BAUET), Qadirabad, Natore, Bangladesh.
- ▷ **3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)** – Research Presenter; University of Dhaka, Dhaka, Bangladesh.

TECHNICAL & RESEARCH SKILLS

Programming:	Python, C, C++, Java, JavaScript, R, SQL
AI / ML:	PyTorch, TensorFlow, Scikit-learn, LightGBM, Deep Learning, NLP, RAG
Computer Vision:	OpenCV, YOLO, FaceNet, ByteTrack, image classification and recognition pipelines
Web / APIs:	Django, React, REST APIs, HTML, CSS, PostgreSQL, SQLite
IoT / Systems:	ESP32, GSM/GPS, embedded and IoT integration, Linux, Docker, Git/GitHub
Research:	Experimental design, dataset preparation, model evaluation, LaTeX, Overleaf, Zotero, Mendeley, academic writing